Supplement to: Characterize Before You Loop: What Transfers From a Zero-Shot Proposer's Template and What Does Not

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Section numbers without an S prefix refer to the main paper. Numbers in this supplement are the numbers in the paper and in the frozen arm reports. The two exceptions are the post hoc clearance re-reads, for arm MU and for arms GM3 and V, each named with its script in S1 and tabled in S2. Facts attributed to the companion submission are cited to it (Anonymous authors, 2026).

S1 Ledger of the arms this paper reports

Every arm reported in this paper, with the section that reports it, its proposer, design, registered test and outcome. Design gives cells × samples per cell. Registered tests are coded P (prediction), F (falsifier), S (secondary prediction) and R (replication rule), each expanded in its arm's section. These are the ledger rows of the eight arms this paper reports, with the section column pointed at this paper. The arms were run once; arms P and P-D are also reported in the companion, and the other six appear there only by citation to this paper.

Table S1: The eight registered arms this paper reports.

Arm	§	Proposer(s)	Design	Registered test	Outcome
MU (conditioning)	4	weak tier	3 parents × 3 N × 15	P-MU1–P-MU4 + F-MU1	keep-or-improve 26/26 under *B_rival*, which entails the 0/26 on-prediction rate there; on-prediction rate 7/28 under *A_anchor*, below P-MU1's registered 50%, so the anchor is not self-stable; F-MU1 is that rule's negation and adds nothing beyond it
L (iterated loop)	5	weak tier	2 N × 2 archives × 6 rounds × 5	L1–L4 + F-L1	L2 holds; L1 and L3 **not met** (L3 ran opposite to the registered direction; post hoc Fisher exact $p = 0.303$, odds ratio 0.375); L4 open: 0/49 conditioned outputs clear the family (largest excess 1.3×10^{-15})

Arm	§	Proposer(s)	Design	Registered test	Outcome
P (pinned path)	6	weak tier, bare API	15 cells × 15	P-P1: $T(k^*, N)$ modal at $\geq$ 5 of the 7 square cells, including $\geq$ 2 of the 4 discriminating ones, and at $\geq$ 4 of the 5 held-out cells; P-P2: modal at $\leq$ 3 of the 7, leaving exactly 4 of 7 unadjudicated, a registration design gap; P-P3: no valid program output at $N = 13$, 21, 31 clears the family argmax (valid at 10^{-9} with excess $> 10^{-6}$); P-P4: at least one clears; F-P1: pooled on-prediction rate over the four discriminating square cells below 40% of valid outputs	validity collapses: 0/105 valid at the square cells, 4/75 held-out; every direct-emission prediction UNSCOREABLE; P-P3 holds (0 of 12 valid programs clear); same-day agent-runtime control 6/6 valid, anchor 4/6 (post hoc)
P-D (runtime component)	6	weak tier, bare API, $N = 13$	2 conditions × 15	P-PD1 vs P-PD2 vs P-PD3 + S-PD1	P-PD1 holds (closed at 27/30 rows by amendment, verdict unchangeable): thinking on 12/14 valid, anchor modal; system line only 0/13; extended thinking is the component
GM (gemini-flash-lite)	7	2nd vendor, 55 first-party + 85 routed rows	7 N × 20	GM-chain bars	UNDERPOWERED (20 valid / 140) under both serving-path accountings
GM2 (gemma, 4k budget)	7	2nd vendor	7 N × 20	GM-chain bars	0/140 parseable; budget confound, diagnosed by GM3
GM3 (gemma, 16k budget)	7	gemma-4-26b	7 N × 20	P-GM3.1–.3 + falsifier	pooled bar met (57.5%) but discriminating cells 11/43 = 26%, below it (post hoc split); misses are the family argmax construction; the structural bar cannot be met at a truncate cell, whose anchor carries a single radius
V (cross-vendor zero-shot)	7	13 free-tier aliases	5 N × 5	floor + V1/V2	2 aliases meet the V1 pooled bar at point estimate (8/12 and 11/18), gemma-4-31b does not (0/15), V2 a point-estimate pass at each evaluable alias (0/4, 0/9, 0/3; every interval includes the bar)

Registrations and commits. Conditioning arm (MU): `arm_mu_preregistration.txt` with `arm_mu_prompts.json`, committed at `2b7d202` before sampling; decision thresholds, power analysis, tie conventions and falsifier all registered. Second-vendor chain (GM, GM2, GM3): `arm_gm_preregistration.txt` at `37b3adb`, `arm_gm2_preregistration.txt` at `3019aab`, `arm_gm3_preregistration.txt` at `c0b5b97`; the serving-path deviation `arm_gm3_amendment1_openrouter.md` at `67cc4a1` was registered before the affected cells were sampled. Cross-vendor arm (V): `arm_v_preregistration.md` at `5444f10`, ancestry-checked by its runners before any sampling, with three amendments each committed before the sampling it governs; the second of them, `arm_v_preregistration_amendment2.md` at `d579797` (2026-08-12), added the three aliases that later cleared the floor. Loop arm (L): registered at `c9645fe` and scored once at `750bb32`. Pinned-path arm (P): registered at `aa473ad` before sampling, with its scoring code committed afterwards at `596b113`. Thinking-budget arm (P-D): `arm_pd_preregistration.txt` at `5554cc0`, pushed before the first call, and closed at 27 of 30 rows by amendment 1 at `1932cc0`.

Ledgers, scorers and reports. Conditioning arm: raw rows verbatim in `arm_mu_collect.jsonl`, scoring `arm_mu_analysis.py`, frozen output `arm_mu_results.txt`; the post hoc clearance recomputation is `arm_mu_ceiling.py`. Loop arm: `arm_l_collect.jsonl`, stepper `arm_l_step.py`, frozen report `arm_l_report.json`. Pinned-path arm: prompts and hashes `arm_p_prompts.json`, runner `arm_p_run.py`, scorer `arm_p_analysis.py`, report `arm_p_report.json`, ledger `arm_p_collect.jsonl` with 225 rows, six refused by the serving path; post hoc diagnostics `arm_p_diag_temperature.jsonl` and `arm_p_diag_runtime.jsonl`, labelled post hoc in their headers. Thinking-budget arm: builder `arm_pd_build.py`, spec `arm_pd_prompts.json`, runner `arm_pd_run.py`, scorer `arm_pd_analysis.py` importing the pinned-path scorer unmodified, ledger `arm_pd_collect.jsonl` with 27 rows plus the refusal rows kept for resume. The post hoc clearance re-read of arms GM3 and V is `arm_ceiling_gm3_v.py`. Second-vendor chain: `arm_gm_gm3_checkpoint.jsonl` carries every row's raw API response verbatim with serving path and provider tagged, scored by `arm_gm3_analysis.py` into `arm_gm3_report.json` with both serving-path accountings. Cross-vendor arm: every invocation appended live to `arm_v_candidates_raw.jsonl`, scored by `arm_v_score.py`, which imports the companion paper's parse-validate-classify pipeline unchanged. That pipeline is `arm_f_repro.py`, the companion paper's own scorer, shipped with this bundle. The companion paper is included as `companion_anonymized.pdf`. One script cited here, `arm_mu_ceiling.py`, lives in the paper repository rather than the corpus root and is shipped under `paper_repo/loop/`.

The independent re-score. `arm_transfer_independent_rescore.py` recounts every cell-level number in this paper from the raw ledgers, with its own parser, its own validity scorer and a linear-programming oracle blind to the recipe, importing nothing from the arms' scoring code; its report is `arm_transfer_independent_rescore.json`. Closed-form values and family argmaxima agree to at most 3.6×10^{-15}, and of 135 cell-level counts 129 agree. The six that differ are these. Three arm L clearance counts at $N = 31$: a proposal sitting on the argmax to within 1.3×10^{-15} counts as clearing under "excess > 0" and not under the registered "excess $> 10^{-6}$", and no verdict moves. Two arm GM validity counts, at $N = 21$ (7 against 6) and $N = 35$ (3 against 2), where a balanced-bracket scan recovers packings from completions the vendor truncated mid-prose. And arm P-D's sampled-row count for D2 (16 against 13), where three rows logged as call errors with no content are counted as sampled by the re-score. The $N = 35$ recount crosses the chain's floor of 3, so that cell becomes scoreable on the re-score's own count, while arm GM's UNDERPOWERED verdict, which rests on 20 valid of 140 across the whole arm, stands either way. The re-score also split arm MU's ceiling buckets as 18 beyond rounding, 9 within it and 61 not exceeding, correcting an earlier 18, 12 and 58; that correction is corrections-ledger item 36 and §4 carries it.

S2 Deviations from preregistration

The rows below are the companion paper's deviation table, restricted to the arms this paper reports. Arm GM2 ran under the chain registration with no deviation of its own and has no row. Arms GM and GM3 each carry one registered serving-path amendment, and the GM3 row also discloses a ledger repair.

Table S2: Deviations from preregistration, for the arms this paper reports.

Prediction	Registered rule	As registered?	Outcome / reason
GM serving path	first-party API only	amendment registered **before resumption** (`arm_gm_amendment1_openrouter.md`, `a075024`, committed 19 s before the first routed row)	55/140 rows first-party, 85/140 via the routed path with the provider logged as the vendor; UNDERPOWERED under both accountings: all rows, 20 valid, 3 scoreable cells, 1 mode match, 3/20 on-prediction; first-party rows alone, 6 valid, 1 scoreable cell, 0 mode matches; the falsifier fires under neither (§7)
GM3 serving path	first-party API only	amendment registered **before resumption** (`arm_gm3_amendment1_openrouter.md`, `67cc4a1`)	41/140 rows via routed path, provider logged; 59.0% / 57.5% dual accounting, no verdict change (§7); ledger repair at `31a7fc4`: two concurrent runners with stale completion snapshots wrote 3 duplicate rows at $N = 31$ (sample indices 6–8), removed from 93 rows to 90 before the arm was completed to 140 unique rows, the pre-repair file kept under a PREDEDUP suffix
Arm V proposer set	3 aliases served through the opencode gateway	expanded 3 → 13 by amendments 1–2, each registered **before the sampling it governs**	verdicts in §7; one of the thirteen aliases is served under a name no vendor attests, and amendment 1 discloses it as unattributed
Arm V transport repair	single pass, `max_tokens` 8192	amendment 3 registered **before resumption** (mechanical: 24,576 budget, retry, censored-row accounting)	877 attempt rows over the 325 registered slots (314 carry rows); floor evaluated per slot (§7)
GM3 argmax-structure check	—	**post hoc**, disclosed (`diagnostics_gm3_rival.py`)	all 27 discriminating-cell packings at the family argmax value carry its two-radius structure (§7); the registered signature is scoped to k^* and cannot see (k^*-1) structure
Arm L (iterated loop)	L1–L4	registered **before sampling** (`c9645fe`), scored once (`750bb32`); four per-lineage bars rather than one pooled bar per cell, not repaired after the fact; one post-sampling scorer correction, affecting only the 10^{-9} clearance test (corrections ledger item 34): the registered configuration stored the family argmax to seven decimals, so a proposal landing exactly on the family argmax at $N = 31$ tested as 3.7×10^{-8} above the rounded number; the scorer now recomputes the argmax in closed form, that proposal agrees with it to 1.3×10^{-15}, float noise, and clears nothing	L2 **confirmed**; L1 **not met** on 13-DIVERSE, a 2–2 tie at $n = 4$ decided against the prediction by the registered tie rule; L3 **not met**, running opposite to the registered direction (Fisher exact $p = 0.303$), the prediction withdrawn rather than reframed; L4 open and answered — 0 of 49 valid conditioned outputs clear the family argmax (§5)
Arm P (pinned path)	P-P1–P-P4 + F-P1	registered **before sampling** (`aa473ad`); post-review motivation disclosed	225 invocations, 219 returning content and 6 refused by the serving path and recorded as such; validity collapsed to 0/105 at the square cells; direct-emission predictions **not evaluable** under the registered floor; P-P3 holds (0/12); F-P1 denominator empty, reported as such (§6)

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Prediction	Registered rule	As registered?	Outcome / reason
Arm P temperature diagnostic	—	**post hoc**, not preregistered	0/6 valid at temperature 0.0, 0.5 and 1.0, which is 0/18 across the three settings (§6)
Arm P runtime control	—	**post hoc**, not preregistered	same prompt through the agent runtime, same day: 6/6 valid, anchor 4/6 (§6)
Arm P-D (runtime component)	P-PD1/P-PD2/P-PD3 + S-PD1	registered **before sampling** (`5554cc0`), after an outside review asked for a diagnostic beyond temperature; run interrupted at 27 of 30 rows (14 D1, 13 D2) when the serving path refused the last three for credit; closed there by amendment 1 (`arm_pd_amendment1_closed_at_27.md`), since no completion could change the verdict	P-PD1 **holds** and the last three rows could not change it: thinking on 12/14 valid, modal $T(4, 13)$; system line 0/13; S-PD1 met (§6)
Arm MU clearance re-read	—	**post hoc**, not preregistered	the clearance question was never registered for arm MU; 88 valid at the primary tolerance, 18 of them above the argmax by at most 1.53×10^{-6} and every one failing the 10^{-9} gate, each decided by overlap, not containment, and in each the excess exceeds the largest single overlap; 13 of the 18 return their argmax parent's rounded sum unchanged (per-row table below); 63 valid at 10^{-9}, four of them above the argmax by at most 2.5×10^{-8}, inside the 5×10^{-8} recorded-sum granularity, none clearing it (§4) (`arm_mu_ceiling.py`)
GM3 discriminating-cell split	—	**post hoc**, not preregistered	the registered bars were the pooled rate, the per-cell modal count and the falsifier; no bar was registered on the discriminating cells alone; read post hoc, those three scoreable cells give 11/43 = 26%, while the two cells where the anchor equals the family argmax hold 35 of the 46 on-prediction packings (§7)
Arm GM3 and arm V clearance re-read	—	**post hoc**, not preregistered	the clearance question was registered for neither arm; GM3: 80 valid at 10^{-6}, 79 at 10^{-9}, 29 of the 79 above the argmax by at most 5.3×10^{-10}, none clearing it; arm V, three evaluable aliases: 45 valid at 10^{-6}, 39 at 10^{-9}, eight of the 39 above by 8.9×10^{-16}, none clearing (§7, S4) (`arm_ceiling_gm3_v.py`)

The eighteen arm MU outputs above the family argmax at 10^{-6}. One row per output, from the arm MU clearance re-read (`arm_mu_ceiling.py`, overlaps recomputed from the verbatim coordinates). Excess is the recorded sum minus the closed-form argmax. Largest overlap is the largest $r_i + r_j - d_{ij}$ over pairs, the quantity the 10^{-9} test fails on. In every row the excess exceeds the largest overlap, so no single pair accounts for it: thirteen rows return the argmax parent's seven-decimal sum unchanged and carry its rounding, the other five sit above or below that parent sum. Sums are recorded at a 5×10^{-8} granularity. These eighteen sit above the argmax by more than that granularity. Nine further rows of the 88 valid at 10^{-6} exceed it only within that granularity, and the remaining 61 do not.

N	Parent	Slot	Recorded sum	Excess over argmax	Largest overlap	Relation to parent sum
13	argmax parent	1	1.7761427	3.25×10^{-7}	1.34×10^{-7}	equals parent sum
13	argmax parent	3	1.7761427	3.25×10^{-7}	1.34×10^{-7}	equals parent sum
13	argmax parent	5	1.7761435	1.13×10^{-6}	3.34×10^{-7}	above parent
13	argmax parent	7	1.7761439	1.53×10^{-6}	4.34×10^{-7}	above parent
13	argmax parent	9	1.7761427	3.25×10^{-7}	1.34×10^{-7}	equals parent sum
13	argmax parent	13	1.7761427	3.25×10^{-7}	1.34×10^{-7}	equals parent sum
21	anchor parent	7	2.2588850	1.52×10^{-6}	3.05×10^{-7}	above parent
21	argmax parent	11	2.2588840	5.24×10^{-7}	1.05×10^{-7}	above parent
31	argmax parent	2	2.7485284	2.63×10^{-7}	4.38×10^{-8}	equals parent sum
31	argmax parent	3	2.7485284	2.63×10^{-7}	4.38×10^{-8}	equals parent sum
31	argmax parent	4	2.7485284	2.63×10^{-7}	4.38×10^{-8}	equals parent sum
31	argmax parent	6	2.7485284	2.63×10^{-7}	4.38×10^{-8}	equals parent sum
31	argmax parent	7	2.7485284	2.63×10^{-7}	4.38×10^{-8}	equals parent sum
31	argmax parent	8	2.7485284	2.63×10^{-7}	4.38×10^{-8}	equals parent sum
31	argmax parent	10	2.7485284	2.63×10^{-7}	4.38×10^{-8}	equals parent sum
31	argmax parent	11	2.7485284	2.63×10^{-7}	4.38×10^{-8}	equals parent sum
31	argmax parent	12	2.7485282	6.26×10^{-8}	3.76×10^{-9}	below parent
31	argmax parent	13	2.7485284	2.63×10^{-7}	4.38×10^{-8}	equals parent sum

S3 Prompts and hashes

A.1 Bare prompt (`{n}` parameterized; the $N = 13$ instance hashes to `32db485bea625ff9f39f4723ebf1a0 1f337559a9e2cf567fb486928f71f7f8df`):

```
Pack exactly {n} non-overlapping circles inside the unit square [0,1]x[0,1]
so that the sum of their radii is as large as possible.
MUST hold: exactly {n} circles; every circle fully inside the unit square
[0,1]x[0,1] (x-r >= 0, x+r <= 1, same for y); no two circles overlap
(distance between centers >= sum of the two radii; touching is allowed).
Do not write or execute code - construct the packing by reasoning alone.
Output ONLY the raw Python list of {n} [x, y, r] lists. No explanation, no
code fences, no other text.
```

A.2 Registered prompt hashes. bare N=13 `32db485b...`, N=17 `8437df75...`, N=21 `a415425b...`, N=31 `a664d003...`, N=35 `3b08c56e...`, N=37 `2c0a8819...`. Arm P registered the N=43 bare hash `1208e7d2...`, which until then had appeared only in the ledger and in no registration. The second-vendor arms and the cross-vendor arm send these same prompts byte-identical, with the SHA-256 restated in the registration and recomputed at call time, aborting on mismatch. The conditioning arm's mutation prompt is hashed in `arm_mu_prompts.json` and the loop arm reuses it verbatim for generations 1 to 5.

A.3 Dispatch wrapper (the runtime arms only, and not part of the hashed task prompt): `Do not use any tools. Your entire final message must be the answer and nothing else.`

A.4 Parser. Completions are parsed with `ast.literal_eval` after stripping code fences. A completion failing to yield a list of N numeric triples is recorded `literal_eval_failed` and scored invalid, with the failure mode logged. Radii must be strictly positive. Containment and pairwise non-overlap are checked at the two registered tolerances.

S4 Per-cell tables for the second-vendor arms

The chain's verdict table. The registered test column is the GM-chain registration, shared by arms GM, GM2 and GM3. Arm V rows report the bar outcome, with the registration's own label named once; both passing intervals span the 50% bar.

Vendor	Model	Serving path	Registered test	Verdict
Google	gemma-4-26b	first-party API, 7 $N \times 20$ (41/140 rows routed under amendment; dual accounting)	pooled $\geq 30\%$; cell-level ≥ 5; falsifier ≥ 4 fails	pooled bar met (57.5%); discriminating cells 11/43 = 26%, **below** the bar (post hoc split, no bar registered on those cells); cell-level short (modal in 2 of the 5 scoreable cells, against a bar of 5; the falsifier did not fire, needing 4 failures); all three misses are the *family argmax construction* (27/43); the structural bar cannot be met at a truncate cell, whose anchor carries a single radius
Cohere	north-mini-code	free-tier alias, 5 $N \times 5$	V1 $\geq$ 50% of valid on-prediction; V2 family argmax, valid at discriminating cells, $\leq$ 10%	V1 bar **MET** at point estimate (8/12 = 67%, [39–86%]; the interval includes the bar), the registration's label TRANSFERS; V2 point-estimate pass (0/4, [0–49%])
OpenAI (open-weight)	gpt-oss-20b	free-tier alias, 5 $N \times 5$	same	V1 bar **MET** at point estimate (11/18 = 61%, [39–80%]; the interval includes the bar); V2 point-estimate pass (0/9, [0–30%])
Google	gemma-4-31b	free-tier alias, 5 $N \times 5$	same	V1 bar **NOT MET** (0/15, [0–20%]), the registration's label DOES-NOT-TRANSFER; V2 point-estimate pass (0/3, [0–56%])
10 further aliases	(NVIDIA, Liquid, Poolside, inclusionAI, others; one unattributed)	free-tier	floor: $\geq$ 10 of 25 invocations valid	BELOW-FLOOR; failure taxonomies logged, no claim in either direction

Arm GM3 per cell. Recomputable from `arm_gm_gm3_checkpoint.jsonl` by `arm_gm3_analysis.py` with no arguments.

N	anchor (4 dp)	Gemma modal sum	mode freq	valid	on-prediction	MODE-MATCH
13	1.6250	1.7761	7/9	9	0	no (misses upward)
17	2.0518	**2.0518**	19/20	20	20	**yes**
21	2.1000	2.2589	9/18	18	8	no (misses upward)
31	2.5833	2.7485	11/16	16	3	no (misses upward)
35	2.9167	**2.9167**	15/15	15	15	**yes**
37	3.0345	—	—	2	0	UNSCOREABLE
43	3.0714	—	—	0	0	UNSCOREABLE

The two non-discriminating cells are $N = 17$ and $N = 35$, and 35 of the 46 on-prediction packings sit in them. The three scoreable discriminating cells are $N = 13$, 21 and 31, where the pooled rate, read post hoc, is 11 of 43. All 20 two-radius signatures come from $N = 17$, the one extend cell among the matches, while $N = 35$'s 15 on-prediction truncations carry no filler to detect.

Clearance of the family argmax (post hoc, disclosed). The rows of arms GM3 and V were rescored by `arm_ceiling_gm3_v.py`, which imports the chain's scorer and carries the closed-form argmax that `arm_mu_ceiling.py` uses. Of arm GM3's 80 valid packings, 79 are valid at 10^{-9}. The best sum at each of its five scoreable cells equals the family argmax to within 10^{-9}: 29 of the 79 sit above it by at most 5.3×10^{-10} (one row at $N = 13$; the rest by under 10^{-14}), so none clears it under the rule of §3. Arm V's three evaluable aliases give 45 valid outputs at 10^{-6}: 18 for gpt-oss-20b, 12 for north-mini-code and 15 for gemma-4-31b. Of those, 39 are valid at 10^{-9}; eight sit above the argmax by 8.9×10^{-16}, all at $N = 35$, and none clears it. The closest at a discriminating cell is gpt-oss-20b, short of the argmax by 2.5×10^{-3} at $N = 31$ and by 1.2×10^{-4} at $N = 37$. Its best at $N = 35$, and north-mini-code's at $N = 17$, equal the argmax to within 10^{-9}.

Arm V per model. The three aliases that cleared the registered floor are in the paper. The other ten finished BELOW-FLOOR and claim nothing in either direction. Their failure taxonomies are transport-dominated for the aliases served through the opencode gateway, and about half transport and half parse failure for two of them. Retries mean the ledger holds far more attempt rows than slots: gemma-4-31b's 25 slots took 190 attempt rows, 165 of them transport failures, and the floor is evaluated on the 25 slots.

S5 Per-cell tables for arms P and P-D, and the template-shape null

The rows below are arms P and P-D per cell. Both submissions report these rows in full (§3); they are tabled here so that this submission can be checked on its own (Anonymous authors, 2026). No value is recomputed.

Arm P per cell. Fifteen invocations per cell. Refused rows are serving-path refusals and are counted invalid at the cell they were issued for. Validity is at 10^{-6}; no cell has a different count at 10^{-9}.

channel	N	sampled	returned	refused	valid	note
square	13	15	15	0	0	all failures overlap
square	17	15	15	0	0	all failures overlap
square	21	15	15	0	0	all failures overlap
square	31	15	15	0	0	all failures overlap
square	35	15	15	0	0	one non-positive radius, rest overlap
square	37	15	15	0	0	one count mismatch, rest overlap
square	43	15	15	0	0	one malformed row, rest overlap
held-out	50	15	15	0	4	all four sum to 2.5, family argmax 3.5295867
held-out	58	15	15	0	0	count mismatches dominate
held-out	62	15	15	0	0	count mismatches dominate
held-out	65	15	15	0	0	count mismatches and overlaps
held-out	75	15	14	1	0	count mismatches and one refusal
code	13	15	13	2	4	below the floor of five
code	21	15	15	0	6	the one code cell above the floor
code	31	15	12	3	2	below the floor of five
total	—	225	219	6	16	0 of 105 square, 4 of 75 held-out, 12 of 45 code

No valid output at any cell clears the family argmax. The code cells were executed under a 10-second timeout, and P-P3 holds on their 12 valid programs.

Arm P-D per cell. One cell, $N = 13$, two conditions, closed at 27 of 30 rows by a registered amendment after the serving path refused the last three for credit. The third row is arm P's same-day agent-runtime control at the same cell, post hoc and not preregistered, scored at 10^{-6} only.

condition	rows	valid at 10^{-6}	valid at 10^{-9}	on the anchor
D1 (thinking on)	14	12	11	5 of 12
D2 (system line only)	13	0	0	—
runtime control (post hoc)	6	6	not logged	4 of 6

Under D1 the median row spends 3160 reasoning tokens, a descriptive count carrying no registered bar, $k = 4$ is modal at 6 of 12, and one output is the family argmax 1.7761424. The anchor $T(4, 13) = 1.625$ accounts for 5 of those 6. The sixth is a $k = 4$ grid at radius 0.125 filling only three of its four rows, twelve circles, with a thirteenth of radius 0.05 at the centre, summing to 1.55 rather than the 1.625 a full truncation gives. That extra radius is neither the anchor's nor the family's tangency filler at $k = 4$, which is 0.0517767, so the row sits on the anchor's grid order without sitting on the anchor, and it is the whole difference between the 5 of 12 on the anchor and the 6 of 12 at $k = 4$. Under D2 every failure is an overlap, which is arm P's collapse. The two D1 failures overlap. The two runtime-control outputs that miss the anchor $T(4, 13) = 1.625$ are a 3×3 grid carrying corner fillers at 1.6144 and a centre-plus-border construction at 1.7500.

Which cells discriminate, and by how much. A cell discriminates when the anchor and the family argmax name different values, so that anchoring and family search can be told apart on it. Inside a trap zone the anchor takes the truncate branch, $T(k^*, N) = N/(2k^*)$, and at every trap cell this paper samples the family argmax is the larger of that and the drop-and-fill value $V(k^*-1, N-(k^*-1)^2)$, so the cell discriminates exactly when the drop-and-fill value is the larger. That two-candidate reading is not general: at $N = 60$ and 61, which no arm samples, a lower order with more fillers ($k = 6$) beats both, which is why the scripts maximize over every admissible order rather than over the two. Over the square cells drop-and-fill wins at $N = 13$, 21, 31 and 43, where the anchor and the argmax are 1.625 against 1.7761424, 2.1 against 2.2588835, 2.5833333 against 2.7485281, and 3.0714286 against 3.2416246. At $N = 35$ the truncated anchor is the larger of the two, and at $N = 17$ and 37 the branch rule extends and the anchor is itself the argmax, so those three cells cannot discriminate.

The template-shape null, derived. The null asks what an on-prediction rate would look like if the proposer picked a template shape rather than anchoring on one. A proposer already inside the recipe family still has to choose a grid order, and the orders in reach at a cell are the nearest-square order k^* and its two neighbours, k^*-1 and k^*+1; an order further out puts the grid radius far from anything the cell can carry. Each order admits one branch value, the extend value where $N \geq k^2$ and the truncate value otherwise, and an extend value is admissible only when its fillers fit the interstices, $N - k^2 \leq (k-1)^2$. At every cell studied but one the three orders are admissible, so the shape choice is a choice among three values, exactly one of which is the anchor, and choosing uniformly lands on-prediction one time in three. At $N = 17$ the order-3 extend value needs 8 fillers against 4 interstices, so that cell has two admissible orders and a null rate of one in two. The rate is therefore per row, and a pooled tail is the exact Poisson-binomial upper tail over the rows' rates; it reduces to Binomial$(n, 1/3)$ wherever no $N = 17$ row is pooled. The rate is a shape-uniform rate inside the recipe family and not a rate over all packings. The companion paper derives the same null, and this paper applies it post hoc, as it does there (Anonymous authors, 2026). Tails are computed by `diagnostics_transfer_null.py` in the artifact.

arm	alias	on-prediction / valid	exact upper tail
V	`north-mini-code`	8/12 pooled (3 rows at $N = 17$)	0.037
V	`gpt-oss-20b`	11/18 pooled	0.014
V	`north-mini-code`	3/4 discriminating cells	0.111
V	`gpt-oss-20b`	5/9 discriminating cells	0.145
GM3	gemma-4-26b	11/43 discriminating cells	0.895

The two pooled rates clear the null and the three discriminating-cell rates do not. Under a flat Binomial$(n, 1/3)$ the north-mini-code pooled tail would read 0.019; the admissibility correction moves it to 0.037 and no verdict.

References

Anonymous authors. Capability or optimizer? what lets an LLM proposer leave its template family on circle packing, 2026. Anonymous companion submission, under review. An anonymized copy accompanies this submission as supplementary material (`companion_anonymized.pdf`).